UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

Algebra/Topology Seminar

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Resolution Transfer and Systems of Higher Homotopies

Thursday, October 22, 2026
3:00 p.m. in Massry B008

Abstract. A common approach to studying the homological properties of a commutative ring S is to transfer resolutions across ring maps $R \to S$. The bar construction provides a universal method for this transfer, however the resolutions obtained are rarely minimal. In this talk, we will introduce systems of higher homotopies on resolutions, inspired by the work of Eisenbud and Shamash. We will show how these systems can be used to transfer minimal resolutions, refining the transfer from the bar construction.